A FINITE TYPE MODULE FLAT ALONG A FIBRE BECOMES FREE AFTER AN ELEMENTARY ETALE NEIGHBOURHOOD

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source line 3623, 'Some minor details omitted.', is retained verbatim at the end of the counted proof. The cover of the fibre, the choice of the dominating neighbourhood and the computation of the global sections of the disjoint union are all carried out inside the span; only the routine verifications of the remaining listed properties are left to the reader.

Allowed standard prerequisites (editorial).

- The pointwise free version of the flat along the fibre lemma. Same chapter flat.tex, label lemma-finite-type-flat-at-point-free, invoked at proof line 3585. Produces, for each point of the fibre, an elementary etale neighbourhood and an affine etale neighbourhood of the point whose global sections of the pulled back module are free over the base, with the residue fields preserved; the passage from finitely many such pieces to one global neighbourhood is what the counted proof supplies.
- Existence of an elementary etale neighbourhood dominating finitely many given ones. More on Morphisms, label more-morphisms-lemma-elementary-etale-neighbourhoods, invoked at proof line 3607. Supplies one elementary etale neighbourhood of the base dominating the finitely many neighbourhoods attached to the chosen points, which is what makes the disjoint union of the pieces into a single neighbourhood.

Source reference. Original lemma 05KY: sources/stacks-flat/pinned-original-source.tex, source lines 3461–3461. Navigation pointer for \ref{lemma-finite-type-flat-at-point-free} used inside the counted span (line 3585); the argument that consumes it is retained in the span.

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Source reference. Original lemma 057B: sources/stacks-chow/tags.txt, source lines 6739–6739. Navigation pointer for \ref{more-morphisms-lemma-elementary-etale-neighbourhoods} used inside the counted span (line 3607); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $s \in S$. Assume that*

(1) *f is of finite presentation,*
(2) *$\mathcal{F}$ is of finite type, and*
(3) *$\mathcal{F}$ is flat over S at all points of X_s.*

Then there exists an elementary étale neighbourhood $(S', s') \to (S, s)$ and a commutative diagram of schemes

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow \\ S & \longleftarrow & \mathrm{Spec}(\mathcal{O}_{S',s'}) \end{array}$$

such that $X' \to X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ is étale, $X_s = g((X')_{s'})$, the scheme X' is affine of finite presentation over $\mathcal{O}_{S',s'}$, the sheaf $g^*\mathcal{F}$ is of finite presentation over $\mathcal{O}_{X'}$, and such that $\Gamma(X', g^*\mathcal{F})$ is a free $\mathcal{O}_{S',s'}$-module.

Proof. For every point $x \in X_s$ we can use Lemma 05KY to find an elementary étale neighbourhood $(S_x, s_x) \to (S, s)$ and a commutative diagram

$$\begin{array}{ccc} (X, x) & \xleftarrow{g_x} & (Y_x, y_x) \\ \downarrow & & \downarrow \\ (S, s) & \longleftarrow & (\mathrm{Spec}(\mathcal{O}_{S_x,s_x}), s_x) \end{array}$$

such that $Y_x \to X \times_S \mathrm{Spec}(\mathcal{O}_{S_x,s_x})$ is étale, $\kappa(x) = \kappa(y_x)$, the scheme Y_x is affine of finite presentation over $\mathcal{O}_{S_x,s_x}$, the sheaf $g_x^*\mathcal{F}$ is of finite presentation over $\mathcal{O}_{Y_x}$, and such that $\Gamma(Y_x, g_x^*\mathcal{F})$ is a free $\mathcal{O}_{S_x,s_x}$-module. In particular $g_x((Y_x)_{s_x})$ is an open neighbourhood of x in X_s. Because X_s is quasi-compact we can find a finite number of points $x_1, \ldots, x_n \in X_s$ such that X_s is the union of the $g_{x_i}((Y_{x_i})_{s_{x_i}})$. Choose an elementary étale neighbourhood $(S', s') \to (S, s)$ which dominates each of the neighbourhoods (S_{x_i}, s_{x_i}), see More on Morphisms, Lemma 057B. Set

$$X' = \coprod Y_{x_i} \times_{\mathrm{Spec}(\mathcal{O}_{S_{x_i},s_{x_i}})} \mathrm{Spec}(\mathcal{O}_{S',s'})$$

and endow it with the obvious morphism $g : X' \to X$. By construction $X_s = g(X'_{s'})$ and

$$\Gamma(X', g^*\mathcal{F}) = \bigoplus \Gamma(Y_{x_i}, g_{x_i}^*\mathcal{F}) \otimes_{\mathcal{O}_{S_{x_i},s_{x_i}}} \mathcal{O}_{S',s'}.$$

This is a free $\mathcal{O}_{S',s'}$-module as a direct sum of base changes of free modules. Some minor details omitted. □